

Voting analysis project plan

Problem in brief

My task for this project is to create a voting analysis application that processes election data to calculate the number of valid voters per party and put it into a bar chart when instructed and analyse the percentage of member gender, List by constituency, list MPs, list parties, list constituency details and constituencies in the region.

The first step is to get the code to introduce the user to some kind of welcome message and output a menu containing options. We also need to make sure classes are in place that represent each row of the CSV file. When the code is run, it should list the seven options, and the user of the software should be able to press a key from 1-7, depending on which filter they intend to use.

Data sources and structure

The application will use data from a CSV file, with each row representing a constituency and containing key information about those constituencies going down each column. The Python `csv.DictReader` module was used to read this file, creating a `Constituency` object for each row. These objects were then stored in a list, forming the core analysis dataset.

Class design

Create a `Constituency` class to model each constituency. The class should have an initialiser that accepts a dictionary containing details about the constituency. The constituencies are stored in a dictionary.

Implement methods within the class:

- `__str__`
to return a string representation that includes the constituency name, party, and region.
- `GetCountry`
to return to the country of the constituency.
- `GetType`
to return the type of constituency.
Etc.

Functionality

The program is designed to analyse valid voter counts by party. It processes all constituencies, totals the valid votes for each party, and presents the results in a clear and simple format. Another key feature is to calculate the gender balance among elected members, displaying the percentage of male and female representatives while handling any missing data.

I plan to allow users to view a list of constituencies either in a basic format, showing the name, party, and region or with detailed information, such as valid votes.

User interaction

The program will feature a menu-driven interface, allowing users to choose from the options mentioned. Each option will be numbered, and users will be prompted to input their choice. The program will handle invalid inputs by displaying appropriate error messages, ensuring a smooth user experience.

To enhance usability, I will try to make the menu so that it will allow users to retry their last action if an invalid option is entered or return to the main menu.

Error management, comments and testing

Error handling will be taken seriously to make the program robust. I will do this by recording errors in GitHub. And even recording the patches as well. I will also make sure to add plenty of comments so people who read my code can understand what's going on.

Overview

By following this structured plan, the application should meet these objectives. I may tweak and alter things along the way if I think something could be better however, the foundations for how the data is managed and the program set up will be the same. Failures and errors will happen along the way, but it will help me learn and shape my code to become even better and more user-friendly.